

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____K. Deeksha Sree(192424097)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Deeksha Sree

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. Assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

- D1 cannot work Night.
- D2 must work before D3 (Morning < Afternoon < Night).
- Only one doctor per shift.
- D3 cannot work Morning.
- Every doctor gets exactly one shift.

Use **Backtracking Search** because it incrementally assigns shifts and backtracks whenever a constraint is violated.

Doctors:

- D1
- D2
- D3

Domains:

Morning (M)

Afternoon (A)

Night (N)

Step 1

Assign

D1 → Morning

Current Assignment

Doctor Shift

D1 Morning

Constraint Check

D1 is not Night

Valid

Step 2

Assign

D2 → Afternoon

Current Assignment

Doctor Shift

D1 Morning

D2 Afternoon

Constraint Check

D2 before D3

Still possible

Step 3

Assign

D3 → Night

Current Assignment

Doctor Shift

D1 Morning

D2 Afternoon

D3 Night

Constraint Check

D3 not Morning

D2 before D3

One doctor per shift
All assigned
Goal reached

Doctor Shift

D1 Morning
D2 Afternoon
D3 Night

Backtracking checks every assignment before moving forward. Since no constraint is violated, the first complete assignment becomes the solution.

2. Robot moves on a 5×5 grid

Movement

- Up
- Down
- Left
- Right

Obstacle cannot be crossed.

Cost per move = 1

Heuristic = Manhattan Distance

Assume Grid

S - - X -

- X - X -

- X - - -

- - X X -

- - - G -

S = Start

G = Goal

X = Obstacle

Use **Breadth First Search (BFS)**

Reason

Since every move costs 1, BFS guarantees the shortest path.

Initially

Queue

[S]

Visited

S

Expand S

Queue

(0,1)

(1,0)

Expand (0,1)

Queue

(1,0)

(0,2)

Expand (1,0)

Queue

(0,2)

(2,0)

Expand (0,2)

Queue

(2,0)

(1,2)

Continue until Goal

Final Path

S

↓

↓

→

→

↓

↓

→

G

Or

(0,0)

↓

(1,0)

↓

(2,0)

→

(2,1)

→

(2,2)

↓

(3,2)

↓

(4,2)

→

Goal

Cost

Moves = 8

Total Cost =

$8 \times 1 = 8$

BFS explores nodes level by level and guarantees the shortest path when all edges have equal cost.

3. Robot moves only

- Up
- Down
- Left
- Right

Decision

Search is limited to four directions.

Blocked cells cannot be entered.

Decision

Robot searches for alternate routes.

Robot only observes neighbouring cells.

Decision

Robot cannot know the whole map initially.

Must explore gradually.

Normal Cost

=1

Risk Zone Cost

=3

Decision

Robot prefers safer routes even if they are slightly longer.

Environment changes while moving.

Decision

Robot updates its map continuously.

(b) Solution Approach

Search Strategy

Use

Uniform Cost Search + Online Search Agent

Steps

Step 1

Start from S



Step 2

Observe neighbouring cells



Step 3

Calculate movement cost



Step 4

Avoid blocked cells



Step 5

Expand lowest-cost path



Step 6

Update knowledge whenever new obstacles appear



Step 7

Repeat until Goal

(c) AI Concepts

Intelligent Agent

The rescue robot is an intelligent agent because it

- senses surroundings
- reasons
- plans
- acts autonomously

Rationality

Robot chooses the safest and cheapest path.

Its objective is

- rescue survivor
- minimize travel cost
- avoid danger

Property	Type
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Observability	Partially Observable
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Agents	Single Agent
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Environment	Dynamic
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Nature	Sequential
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State Space	Discrete
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Knowledge	Unknown
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Uniform Cost Search is suitable because

- movement costs differ
- risky cells have higher cost
- always finds minimum-cost path

Online Search Agent is suitable because

- robot does not know complete environment
- discovers obstacles while moving
- continuously updates decisions

Together they provide

Safe navigation

Lowest travel cost

Dynamic adaptation

Successful rescue